Unsupervised Manifold-Preserving and Weakly Redundant Band Selection Method for Hyperspectral Imagery

Chenhong Sui, *Member, IEEE*, Chang Li, Jie Feng, *Member, IEEE*, and Xiaoguang Mei

***Abstract*— Hyperspectral band selection is of great value to alleviate the curse of dimensionality. For many band selection methods, however, the neglect of bandwise usefulness tends to result in the loss of valuable bands, but the retention of useless ones; consequently, this causes deterioration of the classification performance. In this sense, bandwise significance should be emphasized. To address this issue, this article proposes a manifold-preserving and weakly redundant (MPWR) unsupervised band selection method. In the method, a manifold-preserving band-importance metric is put forward to measure the bandwise essentiality. This ensures the retention of bands involving abundant intrinsic structures conductive to classification. Specifically, aimed at obtaining the presented band-importance metric, an attainment algorithm is presented, which mainly relies on the embedding learning and linear regression, followed by the introduction of multi-normalization combination. In addition, concerning the massive redundancy caused by the highly correlated bands, MPWR further establishes a constrained band-weight optimization model. Then, both bandwise manifold-preserving capability and intraband correlation are fully integrated into the band selection process. To solve the problem, a corresponding algorithm within the framework of the alternating direction method of multipliers (ADMM) is also developed. Regarding evaluating the effectiveness of the proposed method, comparative experiments with the state-of-the-art methods are conducted on three public hyperspectral data sets. Experimental results demonstrate the superiority and robustness of MPWR.**

***Index Terms*— Band-weight optimization, hyperspectral band selection, manifold-preserving band importance, redundancy.**

I. Introduction

HYPERSPECTRAL IMAGES (HSIs) contain hundreds of narrow and continuous spectral bands, which are capable of providing fine spectral reflection characteristics of materials [1]–[3]. The elaborate spectral information, along with the high spatial resolution, makes HSIs extremely potential for precise object recognition and classification [4]. Unfortunately, due to the limitation of insufficient training samples, the high-dimensional spectral bands are accompanied with the curse of dimensionality, which could result in the deterioration of classification performance [5], [6]. Additionally, the increased burdens on data processing, transmission, and storage space are also induced by the large volume of data. Therefore, dimensionality reduction becomes a highly concerned problem for HSI analysis [7]–[10].

The dimensionality reduction techniques can be generally divided into two categories, i.e., feature extraction and band selection. Feature extraction mainly relies on finding an embedding and then projects the original high-dimensional data into a lower dimensional feature space. In such a case, a more abstract representation of the original data is generated, which attempts to keep some properties the same as those in the original space, e.g., the intrinsic manifold structure. For example, the principal component analysis (PCA) endeavor to find the projection directions with maximal variances [11], whereas locally linear embedding (LLE) [12], locality-preserving projections (LPPs) [13], orthogonal LLPs (OLPPs) [14], Laplacian eigenmap (LE) [15], and neighborhood-preserving embedding (NPE) [16] aim at reserving the geometrical structure of data. Different from the above-mentioned methods, the main idea of linear discriminant analysis (LDA) [17] is to seek the effective projection for better discriminating diverse classes.

It is known that these algorithms perform well in terms of reducing dimensionality, while due to the destruction of spectrum structure, they cannot preserve the physical meaning of each band. Correspondingly, the interpretability of substances is severely influenced [18]. By contrast, band selection is devoted to selecting several bands from the original data on the basis of some specific criteria. Thus, the physical significances of spectral signatures can be retained. On this account, band selection is a reasonable alternative for hyperspectral dimensionality reduction.

According to the quantity of supervised information that is utilized, band selection can be grouped into supervised [19]–[21], semisupervised [22]–[24], and unsupervised methods [25]–[27]. Both supervised and semisupervised methods heavily depend on the supervised information to identify

Manuscript received January 25, 2019; revised July 6, 2019; accepted September 11, 2019. Date of publication October 23, 2019; date of current version January 21, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61601397 and Grant 41901350, and in part by the Fundamental Research Funds for the Central Universities under Grant WL15B08, Grant JZ2019HGBZ0151, and Grant JZ2019HGTA0049. *(Corresponding author: Chenhong Sui.)*

C. Sui is with the School of Opto-Electronic Information Science and Technology, Yantai University, Yantai 264005, China (e-mail: sui6662008@163.com).

C. Li is with the Department of Biomedical Engineering, Hefei University of Technology, Hefei 230009, China (e-mail: changli@hfut.edu.cn).

J. Feng is with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: jiefeng0109@163.com).

X. Mei is with the Electronic Information School, Wuhan University, Wuhan 430072, China (e-mail: meixiaoguang@gmail.com).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2944189

0196-2892 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

relevant bands. In these methods, since label information allows for evaluating the class separability, only the bands with strong discriminative information are ultimately selected. Apparently, these methods are more classification-oriented and advantageous to achieve superior classification performance [28]. However, tagging samples is an expensive, tedious, and time-consuming task. In contrast, unsupervised methods require no labeling. Therefore, unsupervised band selection becomes a highly popular research topic.

Based on the strategies, existing unsupervised band selection methods can be mainly separated into three kinds, i.e., ranking-based, searching-based, clustering-based, sparsity-based, and hybrid scheme-based methods [29]–[31]. In terms of the criteria used in band selection, we can then group these methods into the correlation-based [32]–[34], the sparsity-based [21], [25], and information-theoretic-based [35], [36]. It is well known that the correlation-based criteria are beneficial to reduce redundancy. Nonetheless, the bandwise usability is neglected. This implies that valuable or representative bands can be discarded. The sparsity-based methods attempt to employ the nonzero representation coefficients to yield the aimed subset. Regarding the information-theoretic-based methods, they tend to evaluate the amount of information involved in each band. Nevertheless, it is worth noting that these criteria are sensitive to noise and may exert negative effects on the classification performance. For example, the band selection methods on the basis of the entropy would take the very noisy bands as informative ones instead.

To this end, this article proposes a manifold-preserving and weakly redundant (MPWR) unsupervised band selection method. In this method, manifold preservation-based band-importance metric, denoted as $\boldsymbol{v}$, is put forward to depict the bandwise manifold-preserving capability. The rationale behind $\boldsymbol{v}$ comes from the fact that the manifold-preserved bands, involving the intrinsic manifold structures of data, are rich in structural information and valuable to classification [37], [38]. For the sake of attaining $\boldsymbol{v}$, a corresponding band-importance attainment algorithm is developed. In specific, with the employment of embedding learning and linear regression, we can obtain the coefficient matrix, in which the band-specific contributions to retaining the manifold structure are reflected. Then, $\boldsymbol{v}$ can be extracted from the coefficient matrix, followed by the min–max normalization. Additionally, concerning the amplitude discrepancy involved in the matrix, we also impose a column maximum normalization on it. Moreover, to remove the redundancy caused by the adjacent similar bands, $\boldsymbol{v}$ is further integrated with intraband correlation, and a constrained band-weight learning model is then constructed. For solving the problem, this article also presents an optimization algorithm under the popularly used alternating direction method of multipliers (ADMM) [39], [40] framework. Comparative experiments on three broadly used hyperspectral data sets demonstrate the superiority and robustness of the proposed method.

The main contributions of this article can be summarized as follows.

1) A manifold-preserving band-importance metric is proposed in this article. This metric helps us to identify the representative bands with sufficient intrinsic geometrical structural information and is beneficial to classification. Additionally, a band-importance metric attainment algorithm is provided as well.
2) The manifold-preserving band-importance metric is integrated with the intraband correlation, and a corresponding band-weight learning model is further formulated. This ensures that the retained bands can preserve the manifold of HSI and have as less redundancy as possible. Moreover, band selection is ultimately converted to a constrained optimization problem.
3) A corresponding optimization algorithm is developed to obtain the bandwise weight within the framework of ADMM. Comparative experiments with eight state-of-the-art methods demonstrate the superiority and robustness of MPWR.

The rest of this article is organized as follows. In Section II, related unsupervised band selection methods are reviewed. Section III gives a detailed description of the proposed band selection method, including the development and attainment of manifold-preserving band-importance metric $\boldsymbol{v}$, the formulation of MPWR, and the optimization algorithm to tackle MPWR. Section IV chiefly provides the experimental results and discussion. The conclusions and future work are given in Section V.

II. Related Work

To evaluate the bandwise manifold-preserving performance, this article adopts linear embedding to generate the embedded data with the intrinsic manifold structure. Therefore, in this section, a brief introduction related to embedding learning is given. Additionally, for ease of reading, we would like to briefly review some representative unsupervised band selection methods used for comparative analysis in Section IV.

A. Embedding Learning

Let $\boldsymbol{Y} = [\boldsymbol{y_1}, \boldsymbol{y_2}, \ldots, \boldsymbol{y_p}] \in \mathbb{R}^{l\times p}$ be the original high-dimensional data. Then, according to the manifold theory in LLE [12] and NPE [16], we could find a low-dimensional embedded data $\boldsymbol{F} = [\boldsymbol{f_1}, \boldsymbol{f_2}, \ldots, \boldsymbol{f_p}] \in R^{d\times p}$, which preserves the intrinsic geometric properties of $\boldsymbol{Y}$. This means that $\boldsymbol{F}$ also accords with the manifold structure characterized by $\boldsymbol{G}$. In such a case, the manifold preserved low-dimensional embedding $\boldsymbol{F}$ can be obtained by

$$\begin{aligned} &\arg\min_{\boldsymbol{F}} \sum_{m=1}^{p} \left\| \boldsymbol{f_m} - \sum_{n=1}^{p} g_{mn} \boldsymbol{f_n} \right\|_2^2 \\ &\text{s.t. } \boldsymbol{F}\boldsymbol{F}^T = \boldsymbol{I}_{d\times d} \end{aligned} \tag{1}$$

where g_{mn} explicitly illustrates the local neighboring structure between $\boldsymbol{y_m}$ and $\boldsymbol{y_n}$ in $\boldsymbol{Y}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

B. Some Representative Unsupervised Band Selections

In this section, most unsupervised band selection methods associated with comparative experiments will be briefly reviewed. Note that these methods are capable of covering ranking-based, sparsity-based, clustering-based, searching-based, and hybrid scheme-based methods.

Linearly constrained minimum variance-based band correlation constraint (LCMV-BCC) is a ranking-based band selection method [34]. This method takes the BCC into consideration. While different from LCMV-BCC, linearly constrained minimum variance-based band correlation minimization (LCMV-BCM) [34] incorporates the BCM into the band selection process.

For the sparse representation-based method (SpaBS) [25], it attempts to find the sparse representation of the original HSI on the basis of the sparsity theory. As each column of the coefficient matrix can reflect the significance of each basis in constructing the HSI, the bands corresponding to the highest occurrence frequency in the sparse coefficient matrix are collected.

A dominant-set-extraction-based selector (DSEBS) [41] is a graph-based clustering method. In this method, both the bandwise informativeness and independence are quantified based on the extracted structural information. Obviously, this facilitates low computational complexity of DSEBS. Due to the neglect of fine-grained spatial information, weak correlated bands might be characterized as strongly correlated ones instead. In [33], the hierarchical clustering algorithm based on Ward's linkage and divergence information (WaLuDi) is employed to separate the bands into clusters.

Volume-gradient-based band selection (VGBS) [42] is a representative eliminating searching-based method. In VGBS, all bands are treated as the candidate bands and used to construct a parallelotope. Then, VGBS sequentially removes the highly correlated bands based on the gradient of parallelotope volume.

In contrast, hybrid scheme-based methods simultaneously employ multiple schemes for band selection [29]. For example, in [43], the clustering algorithm is used to yield the clusters. After that, the ranking scheme is adopted to find the most dissimilar ones based on the affinity matrix. Similarly, enhanced fast density-peak-based clustering (E-FDPC) [32] employs a ranking-based as well as a clustering-based scheme. In E-FDPC, the ranking score is first captured by weighting the normalized local density and the intracluster distance. Then, E-FDPC can automatically identify the cluster centers and reduce redundancy.

III. Proposed Band Selection Method

Assume that $\boldsymbol{X} = [\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x_p}] \in \mathbb{R}^{l\times p}$ is the original high-dimensional hyperspectral image, where p is the number of pixels, and l represents the number of spectral bands. Owing to the high spatial resolution, HSI has abundant spatial information conductive to classification, e.g., contextual information or geometric structural information of objects. This leads to a great advantage of HSI to discriminate different classes. Nevertheless, not each band has adequate spatial information. Therefore, to guarantee the classification performance, bands involving rich spatial structural information should be retained in the band selection process. Note that manifold structure is capable of reflecting the local geometrical information of data [13], [15], [38]. On this account, to enhance the classification performance, a sensible way is to retain the spectral bands containing an abundant manifold structure. In this rationale, this article proposes a manifold-preserving band-importance metric, denoted as $\boldsymbol{v}$, to collect a band subset full of essential structure information.

A. Presentation of Manifold-Preserving Band Importance

To evaluate the bandwise capability in manifold preservation, the intrinsic manifold structure of $\boldsymbol{X}$ should be extracted in advance. In this connection, this article adopts embedding learning to generate the manifold-reserved embedded data, i.e., the intrinsic manifold structure. Apparently, by associating the embedded data with $\boldsymbol{X}$, the bandwise potential regarding manifold preservation can be deduced. Concerning this point, this article employs the linear regression to connect $\boldsymbol{X}$ with the manifold-reserved embedded data. Consequently, the regression coefficients can directly reveal the bandwise contributions to fitting the embedded data. Furthermore, due to the manifold-maintaining characteristic of the embedded data, the regression coefficients actually imply the band-specific significance in preserving manifold. Accordingly, the objective function is expressed as

$$\begin{aligned} &\arg\min_{\boldsymbol{A},\boldsymbol{Z}} \sum_{m=1}^{p} \left\| \boldsymbol{z_m} - \sum_{n=1}^{p} s_{mn}\boldsymbol{z_n} \right\|_2^2 + \alpha \|\boldsymbol{A}^T\boldsymbol{X} - \boldsymbol{Z}\|_F^2 \\ &\text{s.t. } \boldsymbol{Z}\boldsymbol{Z}^T = \boldsymbol{I}_{d\times d} \end{aligned} \tag{2}$$

where $s_{mn} (m = 1, 2, 3, \ldots, p; n = 1, 2, 3, \ldots, p)$ explicitly characterizes the local neighborhood relationship between $\boldsymbol{x_m}$ and $\boldsymbol{x_n}$ in $\boldsymbol{X}$. $\boldsymbol{Z} = [\boldsymbol{z_1}, \boldsymbol{z_2}, \ldots, \boldsymbol{z_p}] \in R^{d\times p}$ is the embedded data penalized by s_{mn}, and $d \le l$. Thus, the first term implies that if $\boldsymbol{x_m}$ and $\boldsymbol{x_n}$ are close with regard to spectral information, $\boldsymbol{z_n}$ is prone to play a crucial role in approximating $\boldsymbol{z_m}$, and vice versa. Obviously, through the first term, we can obtain the embedded data $\boldsymbol{Z}$ that embodies the essential manifold structure as in $\boldsymbol{X}$. Meanwhile, the second term aims at capturing the regression coefficient matrix $\boldsymbol{A} = [\boldsymbol{a_1}, \boldsymbol{a_2}, \boldsymbol{a_3}, \ldots, \boldsymbol{a_d}] \in R^{l\times d}$. In view of the manifold-preserving properties of $\boldsymbol{Z}$, $\boldsymbol{A}$ virtually depicts the bandwise contribution to constituting the inherent manifold. Suppose that α is a balance parameter, and $\boldsymbol{S}$ acts as the $p \times p$ affinity matrix consisted of s_{mn}. Then, $\boldsymbol{S}$ is a matrix capable of illustrating the statistic and local geometric properties of $\boldsymbol{X}$.

To describe the pixelwise affinity s_{mn}, instead of adopting the popular Heat kernel [14], [44], this article employs the scale-invariant similarity measure [45] given by

$$s_{mn} = \begin{cases} \dfrac{e_{m,k+1} - e_{mn}}{k e_{m,k+1} - \sum_{t=1}^{n} e_{m,t}}, & \boldsymbol{x_n} \in N(\boldsymbol{x_m}) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where k represents the number of neighbors, and $e_{m,k+1}$ is the $(k+1)^{\text{th}}$ smallest Euclidean distance between $\boldsymbol{x}_m$ and the other pixels in $\boldsymbol{X}$. Similarly, $e_{m,t}$ represents the tth smallest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

Euclidean distance between $\boldsymbol{x_m}$ and the other pixels. It is noteworthy that the Euclidean distance between the mth pixel and the nth one is denoted as e_{mn}. Let $N(\boldsymbol{x_m})$ be the k nearest neighbors of $\boldsymbol{x_m}$.

Obviously, (3) provides the similarity definition between $\boldsymbol{x_m}$ and $\boldsymbol{x_n}$ through s_{mn}. If $\boldsymbol{x_n}$ belongs to the k nearest neighbors of $\boldsymbol{x_m}$, then s_{mn} can be directly calculated. Otherwise, when $\boldsymbol{x_n}$ is not one of the k nearest neighbors of $\boldsymbol{x_m}$, then s_{mn} can be relatively low. In this case, (3) sets s_{mn} to 0. Therefore, the local geometric structure of $\boldsymbol{x_m}$ can be explicitly characterized through (3).

In (3), there are at least two obvious advantages over the widely used heat kernel function. First, it can be found that given k, s_{mn} is available. Therefore, compared with the Heat kernel, (3) can effectively avoid the excessive parameter setting. Second, scale invariance is another evident merit of (3). Then, by substituting (3) into (2), the embedded data $\boldsymbol{Z}$ can be obtained, which forms the foundation of acquiring $\boldsymbol{A}$.

Unfolding the second term of (2) in detail, we can find that z_{im} is actually approximated by the inner product of $\boldsymbol{x_m}$ and the regression coefficient vector $\boldsymbol{a_i}$. More specifically, assume $\boldsymbol{a_i} = [a_{1i}, a_{2i}, a_{3i}, \ldots, a_{li}]^T$ and $\boldsymbol{x_m} = [x_{1m}, x_{2m}, x_{3m}, \ldots, x_{lm}]^T$, then we have

$$z_{im} = \sum_{j=1}^{l} (a_{ji} \cdot x_{jm}). \tag{4}$$

Based on (4), it is evident that $\boldsymbol{a_i}$ serves as the ratio of x_{jm} to regress z_{im}. This illustrates that the jth row of $\boldsymbol{A}$ exhibits the contribution of the jth band of $\boldsymbol{X}$ to constructing $\boldsymbol{Z}$. Therefore, to quantify the band-specific importance to maintain the manifold structure, all entries in the corresponding row of $\boldsymbol{A}$ should be comprehensively considered. To this end, a straightforward way is to utilize the summation of each row of $\boldsymbol{A}$ to measure band-specific importance.

However, due to the great magnitude differences in diverse columns of $\boldsymbol{A}$, the rowwise sum tends to be mainly dominated by the columns with large magnitude. In such a case, directly summing each row of $\boldsymbol{A}$ will lead to inappropriate band importance. To alleviate this issue, a column maximum normalization along each column of $\boldsymbol{A}$ is introduced. Concretely, for any column vector $\boldsymbol{a_i} \in \boldsymbol{A}(i = 1, 2, 3, \ldots, d)$, we first normalize it as $\boldsymbol{a_i} = (\boldsymbol{a_i}/\max(\boldsymbol{a_i}))$. After that, the summation of each row of the normalized $\boldsymbol{A}$ can be obtained, denoted as $\boldsymbol{h} = [h_1, h_2, h_3, \ldots, h_l]^T$. Note that the range of interband correlation is usually between 0 and 1. Accordingly, for consistency, we further employ the min–max normalization. Then, band importance is correspondingly normalized between 0 and 1, which is consistent with interband correlation in scope. However, concerning this scope, the min–max normalization also helps to produce a probability-based band importance. Suppose that $\boldsymbol{v} = [v_1, v_2, v_3, \ldots, v_l]^T$ represents the manifold-preserving band importance. Then, for the jth band, v_j can be expressed as follows:

$$v_j = \frac{h_j - \min(\boldsymbol{h})}{\max(\boldsymbol{h}) - \min(\boldsymbol{h})} \tag{5}$$

Algorithm 1 Attainment of Manifold-Preserving Band Importance

Input:
Original HSI $\boldsymbol{X}$, tradeoff parameter α.
Output: Band importance $\boldsymbol{v}$.
1: Calculate the affinity matrix $\boldsymbol{S}$ according to (3).
2: Obtain the Laplacian matrix based on $\boldsymbol{L} = (\boldsymbol{I}_{p\times p} - \boldsymbol{S})^T(\boldsymbol{I}_{p\times p} - \boldsymbol{S})$.
3: Perform eigenvalue decomposition on $\boldsymbol{L} + \alpha\boldsymbol{I} - \alpha\boldsymbol{X}^T(\boldsymbol{X}\boldsymbol{X}^T)^{-1}\boldsymbol{X}$.
4: Attain $\boldsymbol{Z}$ by adopting the d eigenvectors with respect to the d smallest eigenvalues.
5: Compute $\boldsymbol{A}$ on the basis of (7).
6: Get the band importance $\boldsymbol{v}$ by employing (5).
7: **return** $\boldsymbol{v}$.

where $h_j = \sum_{i=1}^{d} (a_{ji})$. It is worth noting that the computation of $\boldsymbol{v}$ heavily relies on the coefficient matrix $\boldsymbol{A}$. On this account, solving $\boldsymbol{A}$ in (2) becomes the key to attain $\boldsymbol{v}$. Regarding this problem, this article presents a corresponding algorithm to attain $\boldsymbol{v}$ in Section III-B.

B. Attainment of $\boldsymbol{v}$

For the first term of (2), by using the Laplacian matrix $\boldsymbol{L}$, it can be converted to a trace-related item [46], [47]. Let $\boldsymbol{L} = (\boldsymbol{I}_{p\times p} - \boldsymbol{S})^T(\boldsymbol{I}_{p\times p} - \boldsymbol{S})$, then we can simplify (2) to the following form:

$$\begin{aligned} &\arg\min_{\boldsymbol{A},\boldsymbol{Z}} tr(\boldsymbol{Z}\boldsymbol{L}\boldsymbol{Z}^T) + \alpha\|\boldsymbol{A}^T\boldsymbol{X} - \boldsymbol{Z}\|_F^2 \\ &\text{s.t. } \boldsymbol{Z}\boldsymbol{Z}^T = \boldsymbol{I}_{d\times d} \end{aligned} \tag{6}$$

where $tr(\cdot)$ represents the trace operator of a matrix. From (6), we can find that A is only related to $\|\boldsymbol{A}^T\boldsymbol{X} - \boldsymbol{Z}\|_F^2$. Then, by setting the derivative of $\|\boldsymbol{A}^T\boldsymbol{X} - \boldsymbol{Z}\|_F^2$ to zero, the analytical solution of $\boldsymbol{A}$ can be expressed as

$$\boldsymbol{A} = (\boldsymbol{X}\boldsymbol{X}^T)^{-1}\boldsymbol{X}\boldsymbol{Z}^T. \tag{7}$$

Note that there exists the equivalence relation between the trace and Frobenius-norm of a matrix in mathematics, e.g., $\|\boldsymbol{A}^T\boldsymbol{X} - \boldsymbol{Z}\|_F^2 = tr((\boldsymbol{A}^T\boldsymbol{X} - \boldsymbol{Z})(\boldsymbol{A}^T\boldsymbol{X} - \boldsymbol{Z})^T)$ [46], [47]. Then, after substituting (7) into (6), an objective function with respect to $\boldsymbol{Z}$ can be written as follows:

$$\begin{aligned} &\arg\min_{\boldsymbol{Z}} tr(\boldsymbol{Z}(\boldsymbol{L} + \alpha\boldsymbol{I} - \alpha\boldsymbol{X}^T(\boldsymbol{X}\boldsymbol{X}^T)^{-1}\boldsymbol{X})\boldsymbol{Z}^T) \\ &\text{s.t. } \boldsymbol{Z}\boldsymbol{Z}^T = \boldsymbol{I}_{d\times d}. \end{aligned} \tag{8}$$

Generally, problem (8) can be solved by the generalized eigenvalue decomposition [48]. Then, we can attain $\boldsymbol{Z}$ by finding the d eigenvectors of $\boldsymbol{L} + \alpha\boldsymbol{I} - \alpha\boldsymbol{X}^T(\boldsymbol{X}\boldsymbol{X}^T)^{-1}\boldsymbol{X}$ corresponding to the d smallest eigenvalues. As a result, with the obtained $\boldsymbol{Z}$, $\boldsymbol{A}$ is obtainable based on (7). Eventually, we can acquire the band importance $\boldsymbol{v}$ via (5).

In summary, the main procedure of band-importance $\boldsymbol{v}$ attainment algorithm can be detailed in Algorithm 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

C. Formulation of the Band-Weight Learning Model

In terms of band selection, $\boldsymbol{v}$ helps to identify the bands with abundant structural information. Due to the existence of redundancy, the interband correlation should be further considered. It is notable that a band with a greater band-importance score should be assigned a higher priority, whereas for a band highly correlated with the others, its priority should be reduced accordingly. Motivated by this consideration, this article proposes an unsupervised MPWR band selection method. In this method, band selection is transformed to a band-weight learning process. Then, whether a band is selected or not simply relies on the ranking of its band weight.

Suppose that $\boldsymbol{w} = [w_1, w_2, w_3, \ldots, w_l]^T$ represents the band weight to be learned. Let $\boldsymbol{C}$ be the interband correlation matrix that describes the proximity among bands. To select b bands from $\boldsymbol{X}$, the corresponding band-weight learning model is formulated as

$$\begin{aligned} &\arg\min_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{C} \boldsymbol{w} - \eta \boldsymbol{v}^T \boldsymbol{w} \\ &\text{s.t.} \sum_{j=1}^{l} w_j = b, \quad \boldsymbol{w} \geq 0. \end{aligned} \tag{9}$$

It is worth noting that the Pearson correlation coefficient is widely used to measure the similarity or affinity owing to its simplicity and effectiveness [49], [50]. For this reason, in (9), $\boldsymbol{C}$ is characterized by the absolute value of the Pearson correlation coefficient. Then, the first term actually depicts the weighted interband correlation. Meanwhile, the weighted band importance is measured by the second term. η denotes the tradeoff parameter. Obviously, $\boldsymbol{w}$ is jointly determined by the redundancy and $\boldsymbol{v}$. Note that the first term improves more quickly than the second one with the increase of b. On this account, aimed at balancing the two terms, η should also be enhanced monotonically related to b. With this consideration in mind, η is designed as

$$\eta = \frac{b}{\rho}(l-1) \cdot e^{-\frac{b}{l}} \tag{10}$$

where only ρ is the undetermined parameter with $\rho > 0$.

Since the relationship between b and l satisfies $b \leq l$, we have $\partial \eta / \partial b \geq 0$. This means that η is positively related to b. In other words, as the selected number of bands, i.e., b, increases, η is increased accordingly, based on (10). In such a case, η is capable of keeping the balance between the weighted band importance and redundancy automatically.

D. Optimization of MPWR

Aimed at finding the solution of problem (9), this article presents an optimization algorithm within the framework of the ADMM. In this algorithm, considering the nonnegativity of $\boldsymbol{w}$, we introduce the auxiliary variable $\boldsymbol{q}$ with $\boldsymbol{q} = [q_1, q_2, q_3, \ldots, q_l]^T$ and then employ the indicator function. As a result, problem (9) can be transformed to the form without nonnegative constraint given by

$$\begin{aligned} &\arg\min_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{C} \boldsymbol{w} - \eta \boldsymbol{v}^T \boldsymbol{w} + \ell_{\Re_+}(\boldsymbol{q}) \\ &\text{s.t. } \boldsymbol{q} = \boldsymbol{w} \\ &\quad \sum_{j=1}^{l} q_j = b \end{aligned} \tag{11}$$

where $\ell_{\Re_+}(\boldsymbol{q}) = \sum_{j=1}^{l} \ell_{\Re_+}(q_j)$ is the indicator function for the nonnegative orthant $\Re_+$. $l_{R_+}(q_j)$ represents that if the jth element of $\boldsymbol{q}$ belongs to the nonnegative orthant R_+, $l_{R_+}(q_j) = 1$, and $l_{R_+}(q_j) = +\infty$ otherwise. Namely

$$l_{R_+}(q_j) = \begin{cases} 1, & q_j > 0 \\ +\infty, & q_j \leq 0. \end{cases} \tag{12}$$

To acquire the solution of $\boldsymbol{w}$, we further give the augmented Lagrangian form [51] of problem (11) as follows:

$$\begin{aligned} &\arg\min_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{C} \boldsymbol{w} - \eta \boldsymbol{v}^T \boldsymbol{w} + \ell_{\Re_+}(\boldsymbol{q}) + \frac{\mu}{2} \|\boldsymbol{w} - \boldsymbol{q} - \boldsymbol{r}\|_F^2 \\ &\text{s.t.} \sum_{j=1}^{l} q_j = b \end{aligned} \tag{13}$$

where μ represents the Lagrangian multiplier, and $\boldsymbol{r}$ denotes the residual error between $\boldsymbol{w}$ and $\boldsymbol{q}$.

Since problem (13) is separable, we can obtain $\boldsymbol{w}$ by fixing $\boldsymbol{q}$. Then, the objective function with respect to $\boldsymbol{w}$ is

$$L(\boldsymbol{w}) = \arg\min_{\boldsymbol{w}} \boldsymbol{w}^T \boldsymbol{C} \boldsymbol{w} - \eta \boldsymbol{v}^T \boldsymbol{w} + \frac{\mu}{2} \|\boldsymbol{w} - \boldsymbol{q} - \boldsymbol{r}\|_F^2. \tag{14}$$

After setting the derivative $\partial L(\boldsymbol{w}) / \partial \boldsymbol{w}$ to zero, we have the updating rule of $\boldsymbol{w}$ as follows:

$$\boldsymbol{w}^{t+1} = (2\boldsymbol{C} + \mu \boldsymbol{I}_{l \times l})^{-1} \cdot (\eta \boldsymbol{v}^{t+1} + \mu \boldsymbol{q}^t + \mu \boldsymbol{r}^t) \tag{15}$$

where $\boldsymbol{I}_{l \times l}$ is the identity matrix of size $l \times l$.

Analogously, by fixing $\boldsymbol{w}$, we can get the objective function respecting $\boldsymbol{q}$ depicted as

$$\begin{aligned} L(\boldsymbol{q}) = &\arg\min_{\boldsymbol{q}_1} \ell_{\Re_+}(\boldsymbol{q}) + \frac{\mu}{2} \|\boldsymbol{w} - \boldsymbol{q} - \boldsymbol{r}\|_2^2 \\ &\text{s.t.} \sum_{j=1}^{l} q_j = b. \end{aligned} \tag{16}$$

Let the derivative $\partial L(\boldsymbol{q}) / \partial \boldsymbol{q}$ be zero, and we can acquire the iterative rule of $\boldsymbol{q}$ shown as

$$\boldsymbol{q}^{t+1} = \frac{\max(\boldsymbol{0}, \boldsymbol{w}^{t+1} - \boldsymbol{r}^t)}{\sum_{j=1}^{l} \max(0, {w_j}^{t+1} - {r_j}^t)} \cdot b. \tag{17}$$

Then, with $\boldsymbol{w}^{t+1}$ and $\boldsymbol{q}^{t+1}$ fixed, the updating rule associated with $\boldsymbol{r}$ is written as

$$\boldsymbol{r}^{t+1} = \boldsymbol{r}^t - (\boldsymbol{w}^{t+1} - \boldsymbol{q}^{t+1}). \tag{18}$$

Apparently, through the alternate iteration of $\boldsymbol{w}$ and $\boldsymbol{q}$ and also continuous updates of $\boldsymbol{r}$, we can learn the bandwise weight $\boldsymbol{w}$. As $\boldsymbol{w}$ is a comprehensive evaluation of each band in terms of band importance and redundancy, it is used as the criterion to select bands. On this account, bands are ranked based on $\boldsymbol{w}$. Hence, the finally selected b bands are those in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 MPWR

Input:

Band importance $\boldsymbol{v}$, intercorrelation coefficient matrix C, maximum number of iterations φ, tolerance value τ, Lagrangian multiplier μ, tradeoff parameter η.

Output: Band subset of b bands.

1: **Initialization:** $t = 0$.
2: **repeat**
3: $\boldsymbol{w}^{t+1} \leftarrow (2\boldsymbol{C} + \mu \boldsymbol{I}_{l\times l})^{-1} \cdot (\eta \boldsymbol{v}^{t+1} + \mu \boldsymbol{q}^t + \mu \boldsymbol{r}^t)$;
4: $\boldsymbol{q}^{t+1} \leftarrow \frac{\max(\boldsymbol{0}, \boldsymbol{w}^{t+1} - \boldsymbol{r}^t)}{\sum_{j=1}^{l} \max(0, w_j{}^{t+1} - r_j{}^t)} \cdot b$;
5: $\boldsymbol{r}^{t+1} \leftarrow \boldsymbol{r}^t - (\boldsymbol{w}^{t+1} - \boldsymbol{q}^{t+1})$;
6: Update iteration: $t \leftarrow t + 1$.
7: **until** $[(\boldsymbol{w}^{t+1})^T \boldsymbol{C}(\boldsymbol{w}^{t+1}) - \eta \boldsymbol{v}^T \boldsymbol{w}^{t+1}] - [(\boldsymbol{w}^t)^T \boldsymbol{C}(\boldsymbol{w}^t) - \eta \boldsymbol{v}^T \boldsymbol{w}^t] \le \tau$ and $t \le \varphi$.
8: Sort $\boldsymbol{w}$ in descending order.
9: **return** b selected bands with respect to the b largest $\boldsymbol{w}$.

accordance with the b greatest weights. Algorithm 2 gives the main procedures of MPWR in detail.

Therefore, the main process of the proposed band selection method can be summarized as follows. First, we can employ the presented manifold-preserving band-importance estimation algorithm to attain $\boldsymbol{v}$. Assume that b is the number of bands to be selected. Then, given $\boldsymbol{v}$ and b, we can ultimately get the b selected bands through Algorithm 2.

IV. Experiments

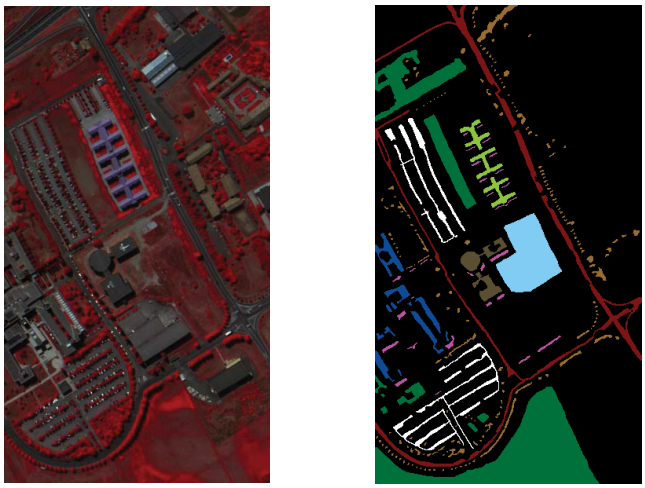
In this section, aimed at evaluating the performance of the proposed band selection method (MPWR), experiments in comparison with several state-of-the-art methods are conducted. These experiments adopt the popular overall accuracy (OA) and kappa statistic coefficient (κ) to assess the effectiveness of each method. Additionally, to compare the effectiveness of different methods on the whole, the average OA (AOA) is also involved. Moreover, considering the statistical significance of classification results, all the OA, κ, and AOA values provided in the following experiments are averaged over ten runs.

We organize the rest of this section as follows. First, in Section IV-A, three popular hyperspectral data sets are described in detail. Second, Section IV-B provides brief introduction about several well-known unsupervised band selection methods used for comparison. Third, the classification algorithms used to evaluate the effectiveness of MPWR are briefly given in Section IV-C. Fourth, to analyze the impact of parameters, Section IV-D gives the parameter analysis of MPWR. Then, to compare the performance of these band selection methods, Section IV-E demonstrates comparisons of classification accuracies with respective to three popular hyperspectral data sets. However, aimed at comparing the performance of each method as a whole, Section IV-F depicts the average OA values related to nine band selection methods. Finally, the space complexity analysis and running times are shown in Section IV-G, accordingly.



(a) (b)

Fig. 1. Information of the AVIRIS Indian Pines image. (a) Pseudocolor image. (b) Ground truth.



(a) (b)

Fig. 2. Information of the ROSIS Pavia University image. (a) Pseudocolor image. (b) Ground truth.

A. Hyperspectral Data Sets

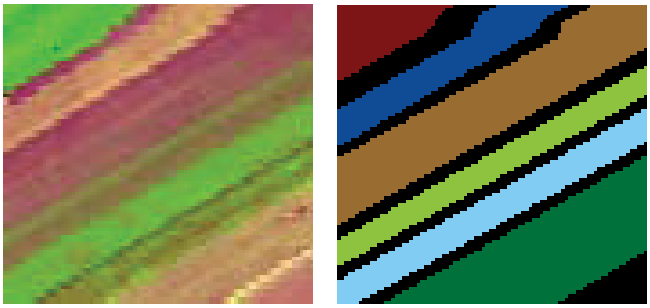
For the sake of demonstrating the effectiveness of MPWR, three widely used hyperspectral data sets, i.e., AVIRIS Indian Pines image, ROSIS Pavia University image, and AVIRIS Salinas-A image, are utilized in this article.

1) AVIRIS Indian Pines Image: The Indian Pines image is about the scene of Northwestern Indiana acquired by the AVIRIS sensor in 1992. Its spatial resolution is 3.7 m with the size of 145 × 145. There are 224 spectral reflectance bands in the range from 0.4 to 2.5 μm. After removing the water absorption bands, i.e., [104-108], [150-163], and 220, only 204 bands are left. Fig. 1 shows the corresponding pseudocolor image as well as the ground truth containing 16 classes.

2) ROSIS Pavia University Image: The Pavia University image is collected by the ROSIS-3 optical sensor over Pavia University. It contains 103 spectral bands ranging from 0.43 to 0.86 nm. Its spatial resolution is about 1.3 m with the size of 610 × 340. In the image, there are nine labeled classes, i.e., Asphalt, Meadows, Gravel, Trees, Painted metal sheets, Bare Soil, Bitumen, Self-Blocking Bricks, and Shadows. The pseudocolor image and the ground-truth are shown in Fig. 2.

3) AVIRIS Salinas-A Image: Salinas-A is a famous subscene of Salinas image gathered by the 224-band AVIRIS sensor over Salinas Valley, CA, USA. It is an 86 × 83 pixel image. Its spatial resolution is 3.7 m. As with the Indian Pines image, 20 water absorption bands are discarded:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 3. Information of the AVIRIS Salinas-A image. (a) Pseudocolor image. (b) Ground truth.

[108-112], [154-167], and 224. This image includes 16 classes. Fig. 3 gives its pseudocolor image along with its ground truth.

B. Description of Compared Unsupervised Band Selection Methods

Aimed at assessing the performance of the proposed MPWR, eight representative unsupervised band selection methods, i.e., E-FDPC [32], SpaBS [25], VGBS [42], WaLuDi [33], LCMV-BCC [34], LCMV-BCM [34], DSEBS [41], and TRC-OC-FPDC [29] are utilized for comparison. According to Section II, eight comparative methods cover ranking-based, clustering-based, searching-based, sparsity-based, and hybrid scheme-based unsupervised band selection methods.

Apart from the eight band selection methods, we also analyze the classification performance on the basis of all bands for comprehensive comparison.

C. Description of Classifier and Experimental Setup

Since the support vector machine (SVM) [52]–[55] is quite advantageous to deal with classification of small samples, it is widely applied to yield the performance comparison of different band selection methods [30], [41], [56]. Therefore, we utilize the popular classification algorithm, SVM, as a classifier. Besides SVM, as in [29], LDA [57] and edge-preserving filtering (EPF)-based spectral–spatial classification algorithm [58] are also employed.

In this article, all the parameters involved in the classifiers or the band selection methods are determined via the popular fivefold cross-validation method or provided by the author. Specifically, in the following experiments, through fivefold cross validation, k is set to 9, 13, and 4 on Indian Pines, Pavia University, and Salinas-A images, respectively. Meanwhile, the values of parameters C and γ in SVM correspond to 5000 and 0.1 on the Indian Pines image, whereas on Pavia University, C and γ are set to 5000 and 4. Respecting the Salinas-A image, they are set as 12000 and 0.7.

D. Impact of Parameters

To assess the effect of parameters on classification, we first randomly select around 10% data as the training data and leave the remaining around 90% for testing. Tables I–III show the number of training and testing data corresponding to the above-mentioned three data sets. In such a case, through SVM, LDA, and EPF, the classification accuracies with the variation of b under different choices of parameters can be acquired. For simplicity, we take the average of the OA values produced by SVM, LDA, and EPF for consideration.

TABLE I

16 CLASSES OF THE INDIAN PINES IMAGE AND THE NUMBERS OF TRAINING AND TESTING SAMPLES FOR EACH CLASS

Class		Number of samples	
Class	Name	Train	Test
2	Alfalfa	5	41
3	Corn-notill	143	1285
4	Corn-mintill	83	747
5	Corn	24	213
6	Grass-pasture	48	435
7	Grass-trees	73	657
8	Grass-pasture-mowed	3	25
9	Hay-windrowed	48	430
10	Oats	2	18
11	Soybean-notill	97	875
12	Soybean-mintill	246	2209
13	Soybean-clean	59	534
14	Wheat	21	184
1	Woods	127	1138
15	Buildings-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	9	84
Total number		1027	9222

TABLE II

NINE CLASSES OF THE PAVIA UNIVERSITY IMAGE AND THE NUMBERS OF TRAINING AND TESTING SAMPLES FOR EACH CLASS

Class		Number of samples	
Class	Name	Train	Test
1	Asphalt	714	5917
2	Meadows	1971	16678
3	Gravel	217	1882
4	Trees	307	2757
5	Painted metal sheets	144	1201
6	Bare Soil	547	4482
7	Bitumen	137	1193
8	Self-Blocking Bricks	371	3311
9	Shadows	92	855
Total number		4500	38276

TABLE III

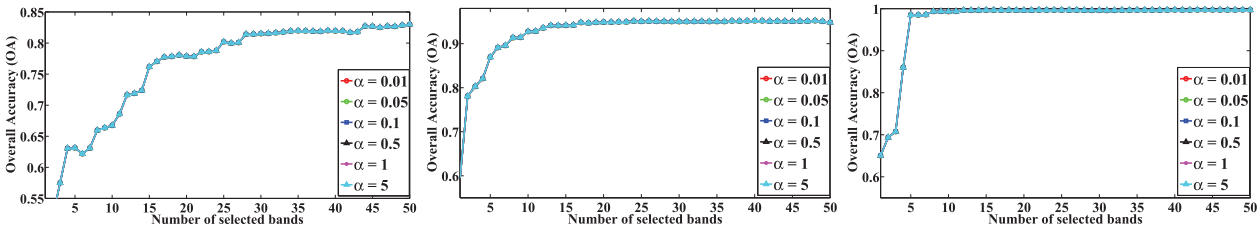
SIX CLASSES OF THE SALINAS-A IMAGE AND THE NUMBERS OF TRAINING AND TESTING SAMPLES FOR EACH CLASS

Class		Number of samples	
Class	Name	Train	Test
1	Brocoli_green_geeds_1	43	348
2	Corn_senesced_green_weeds	129	1214
3	Lettuce_romaine_4wk	50	566
4	Lettuce_romaine_5wk	164	1361
5	Lettuce_romaine_6wk	72	602
6	Lettuce_romaine_7wk	92	707
Total number		550	4798

1) Effect From the Parameter α on the Sensitivity of Classification Accuracy: Equation (6) illustrates that α is a valuable balancing parameter. In this connection, under different numbers of selected bands b, the impact of α in MPWR on the classification accuracy should also be analyzed. Therefore, we set the ranges of α as $\{0.01, 0.05, 0.1, 0.5, 1, 5\}$. The reason for not setting α at a certain step interval is to avoid too many candidate α values [20].

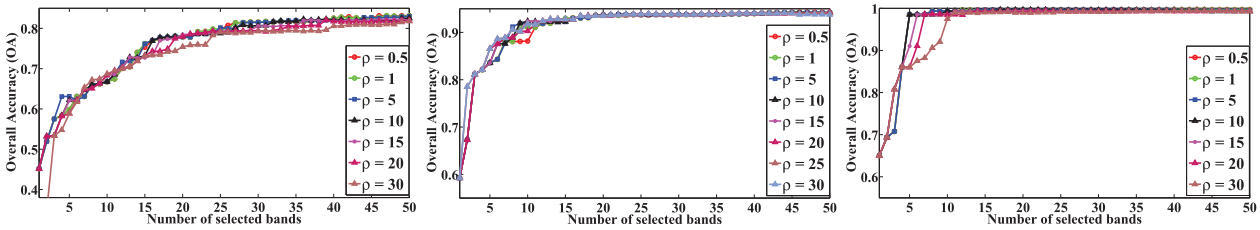
Fig. 4 shows the curves of the classification accuracies against b under different settings of α. In specific, Fig. 4(a)–(c)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.



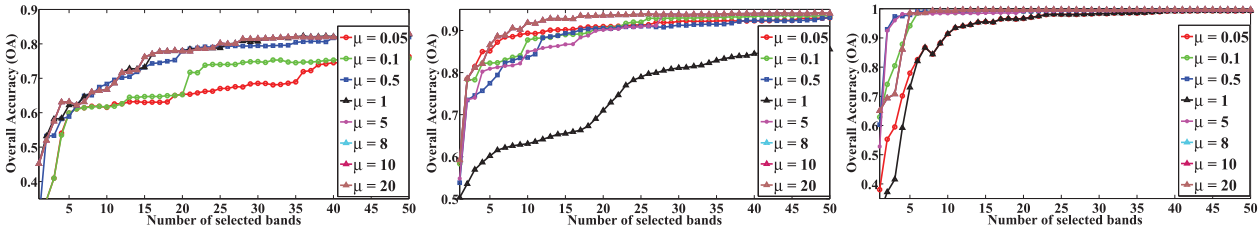
(a) (b) (c)

Fig. 4. Curves of classification accuracy versus the number of selected bands under different choices of α on three images. (a) Indian Pines image. (b) Pavia University image. (c) Salinas-A image. The y-axis represents the average of OA values produced by SVM, LDA, and EPF.



(a) (b) (c)

Fig. 5. Curves of classification accuracy versus the number of selected bands under different choices of ρ on three images. (a) Indian Pines image. (b) Pavia University image. (c) Salinas-A image. The y-axis represents the average of OA values produced by SVM, LDA, and EPF.



(a) (b) (c)

Fig. 6. Curves of classification accuracy versus the number of selected bands under different choices of μ on three images. (a) Indian Pines image. (b) Pavia University image. (c) Salinas-A image. The y-axis represents the average of OA values produced by SVM, LDA, and EPF.

shows the experimental results with regard to Indian Pines, Pavia University, and Salinas-A images, respectively. In Fig. 4, the x-axis represents b, whereas the y-axis denotes the mean of the OA values produced by SVM, LDA, and EPF.

As shown in Fig. 4, the curves of different α values are almost identical on the three images. This indicates that MPWR is not sensitive to α. Therefore, we can set α as 0.1 or other values.

2) Effect From the Parameter ρ on the Sensitivity of Classification Accuracy: Equation (11) reveals that η is an important tradeoff parameter in producing the band-weight vector $\boldsymbol{w}$. In this regard, it is necessary to analyze the influence of η on the classification accuracy. According to (10), η mainly relies on the parameter ρ. Therefore, under different numbers of selected bands b, relevant experiments are conducted to investigate the impact of ρ and then determine the appropriate setting of ρ.

In specific, we set the ranges of ρ as {0.5, 1, 5, 10, 15, 20, 30} on both the Indian Pines and the Salinas-A images. Meanwhile, the range of ρ on Pavia University image is set as {0.5, 1, 5, 10, 15, 20, 25, 30}. Fig. 5 plots the curves of the classification accuracies versus b under different settings of ρ. Analogous to Fig. 4, Fig. 5(a)–(c) corresponds to the experimental results on Indian Pines, Pavia University, and Salinas-A images, respectively. However, the x-axis of Fig. 5 is b, whereas the y-axis describes the average of the OA values produced by SVM, LDA, and EPF.

As shown in Fig. 5, there only exist minor differences between these curves on each image. In this connection, it can be inferred that MPWR is not quite sensitive to the parameter ρ. Despite the high similarity between the classification accuracy curves, we can still find the optimal value of ρ. For instance, in terms of the Indian Pines, Pavia University, and Salinas-A images, the optimal settings of ρ are 5, 25, and 1, respectively.

3) Effect From the Parameter μ on the Sensitivity of Classification Accuracy: Equation (13) shows that μ is crucial in promoting the approximation of $\boldsymbol{w}$ and $\boldsymbol{q}$. On this account, the effect of μ on the classification accuracy for varying b should also be studied. Specifically, instead of setting μ at a certain step interval [20], we set the ranges of μ as {0.05, 0.1, 0.5, 1, 5, 8, 10, 20}. Obviously, this is beneficial to reduce the number of candidate μ values. Fig. 6 depicts the curves of the classification accuracies against b under different settings of μ. Similar to Fig. 5, the x-axis of Fig. 6 denotes b, whereas the y-axis is the mean of the OA values produced by SVM, LDA, and EPF.

Fig. 6(a) and (b) evidently demonstrates that for μ equal to 8, 10, and 20, all the curves coincide. Moreover, these curves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

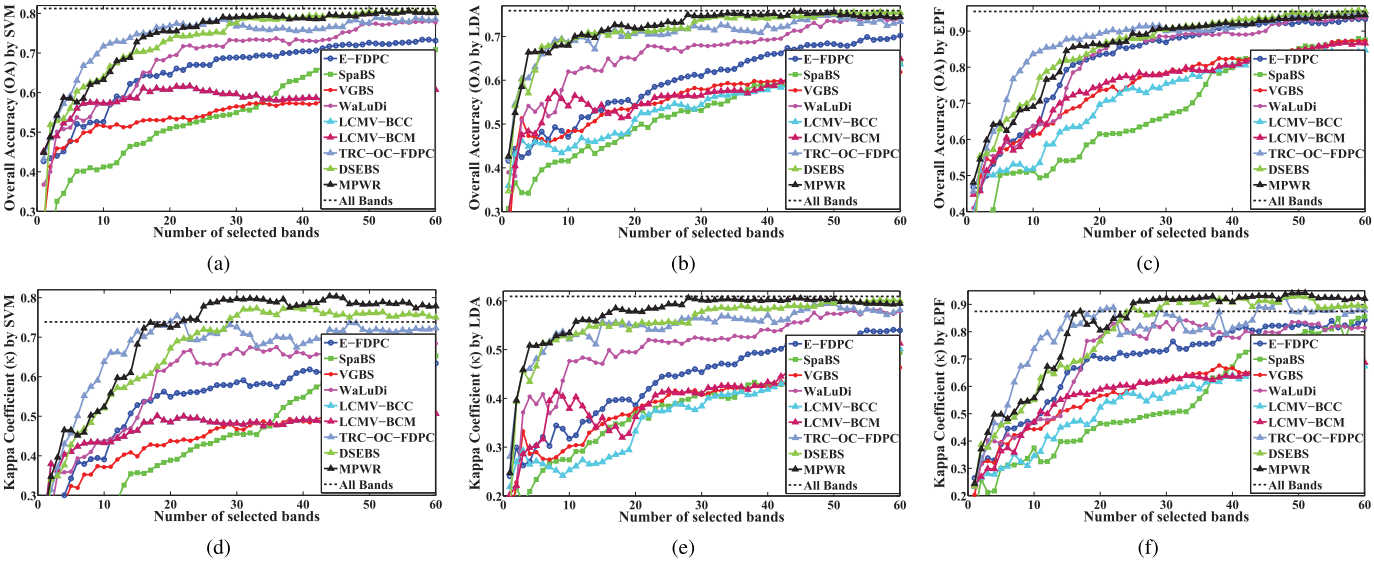


Fig. 7. Classification accuracy curves of different band selection methods on the Indian Pines image with different number of selected bands. OA obtained by (a) SVM, (b) LDA, and (c) EPF. κ obtained by (d) SVM, (e) LDA, and (f) EPF. The black dotted lines represent that all bands are used for classification without selection.

exhibit the highest OA values than others as a whole. Meanwhile, for b greater than 5 in Fig. 6(c), setting μ to 8, 10, or 20 also leads to the best classification performance. Consequently, 8, 10, or 20 are the best candidates for μ.

E. Experimental Results on Indian Pines Image

To evaluate the effectiveness of MPWR, this section mainly focuses on providing the comparative experiments on the Indian Pines image. In the experiments, around 10% pixels of the Indian Pines image are randomly selected for training, and the remaining 90% are used for testing, as shown in Table I. Then, through SVM, LDA, and EPF, we can attain the classification accuracies with respect to each band selection method. For illustrative purpose, the curves of classification accuracy versus b of diverse band selection methods are shown in Fig. 7.

In Fig. 7, the x-axis represents the number of selected bands b, whereas the y-axis characterizes the classification accuracy. Concretely, Fig. 7(a)–(c) plots the SVM-based, LDA-based, and EPF-based OA against b for each band selection method, respectively. Accordingly, the curves regarding κ are shown in Fig. 7(d)–(f). Note that when b approaches 60, the accuracies of most band selection methods are relatively stable, and close to the ones on the basis of all the bands. Therefore, b ranges from 1 to 60 on this image.

As shown in Fig. 7, with the increase of b, there is a general uptrend in terms of both OA and κ for nine methods. Moreover, with b being less than 20, the fastest rise of accuracy usually occurs here. This is largely due to the fact that with small number of bands, the increase of b implies more discriminative information. Then, it is natural that more bands contribute greater to classification.

On this image, VGBS has comparable performance with LCMV-BCC but inferior to WaLuDi. Though WaLuDi is excellent, DSEBS, TRC-OC-FDPC, and MPWR still have obvious advantage over it. It is worth noting that the positions for b being 5 and 25 are two important points. In particular, for b less than 5, MPWR generally exhibits the best performance than others. When b ranges between 6 and 25, it is evident that TRC-OC-FDPC outperforms others under the SVM and EPF classifiers. While as shown in Fig. 7(a) and (d), for b greater than 25, MPWR is almost always superior to the others.

From Fig. 7(b) and (e), we can observe that under the classifier LDA, MPWR gives stable and the best overall performance. Take b for 25 as one example. The corresponding OA and κ of MPWR reach 0.7329 and 0.5931, respectively. Meanwhile, in terms of SpaBS, the OA and κ are 0.5156 and 0.3761. Additionally, what we should not ignore is that DSEBS and TRC-OC-FDPC also achieve satisfactory performance. They outperform others except MPWR.

Similarly, Fig. 7(a)–(f) shows smooth rising curve against b. Note that compared with SVM and LDA, the classification accuracies under EPF are evidently improved. Consequently, E-FDPC and WaLuDi also gain good classification performance. In spite of this, for b less than 5, the best performance is obviously achieved by MPWR. Meanwhile, as shown in Fig. 7(f), for b locating between 6 and 25, TRC-OC-FDPF generally provides the highest κ values. When b is greater than 25, though DSEBS is excellent, which yields the suboptimal performance, MPWR outperforms others in view of classification accuracies.

To give a quantitative comparison, the detailed classification accuracies of each method in the case of 25 selected bands are illustrated in Table IV.

F. Experimental Results on Pavia University Image

Concerning assessing the effectiveness of MPWR, this section presents the comparative experimental results of different band selection methods on the Pavia University image. Like the experiments on the Indian Pines image, about 10% of the Pavia University image are randomly selected for training. Meanwhile, the remaining 90% are used for testing, as described in Table II. Then, we can obtain the SVM-based, LDA-based, and EPF-based classification results related to selected bands obtained by each method. To better quantify the classification results, two indexes, i.e., OA and κ are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

OA AND κ STATISTIC (IN PERCENTAGE) OBTAINED BY SVM, LDA, AND EPF FOR THE INDIAN PINES IMAGE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Classifier	SVM								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	67.00	52.98	54.04	71.56	60.38	60.38	77.14	74.96	**77.97**
κ	56.49	42.96	44.89	63.36	48.96	48.96	69.67	71.96	**77.82**
Classifier	LDA								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	59.46	51.56	55.78	66.92	53.85	56.40	70.54	71.82	**73.29**
κ	44.80	38.40	40.74	51.41	37.61	41.59	54.62	55.69	**59.31**
Classifier	EPF								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	85.70	62.44	75.74	88.02	74.19	76.98	**90.52**	87.90	89.32
κ	72.10	47.43	59.59	79.21	57.30	59.73	79.93	88.71	**90.52**

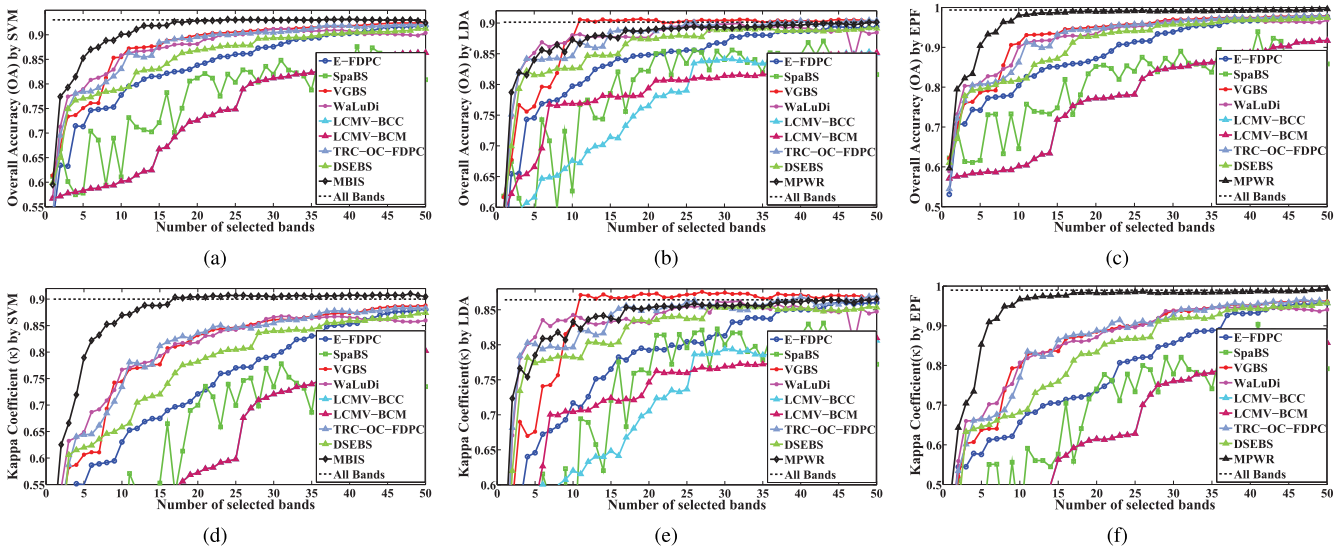


Fig. 8. Classification accuracy curves of different band selection methods on the Pavia University image with different number of selected bands. OA obtained by (a) SVM, (b) LDA, and (c) EPF. κ obtained by (d) SVM, (e) LDA, and (f) EPF. The black dotted lines represent that all bands are used for classification without selection.

involved. The specific experimental results are demonstrated in Fig. 8.

In Fig. 8, the x-axis represents the number of selected bands b, whereas the y-axis is the classification accuracy. Concretely, Fig. 8(a) and (d) plots the SVM-based accuracy curves respecting the two indexes, i.e., OA and κ, while the LDA-based accuracy curves are shown in Fig. 8(b) and (e). Accordingly, Fig. 8(a) and (d) shows the EPF-based accuracy curves. On this image, it is worth noting that as b reaches about 45, the classification accuracies tend to be relatively stable and approximate the ones of all the bands. Therefore, in Fig. 8, the accuracy curves are plotted with b ranging from 1 to 50.

We can observe from Fig. 8 that, for b below 20, aside from SpaBS, all the other eight methods provide greater classification accuracies with the increase of b as a whole. This is mainly due to the fact that when b is small, the discriminative information is not adequate. In this case, the increase in b can guarantee more information, which contributes greatly to classification. For instance, when b increases from 1 to 2, the curves of MPWR exhibits the fastest upward trend. Specially, in Fig. 8(a), when b ranges from 1 to 2, the OA values of MPWR changes from 0.5948 to 0.7741. Apparently, an improvement of nearly 0.18 is yielded.

From Fig. 8(a) and (d), we can find that the performance of VGBS, WaLuDi, and TRC-OC-FDPC are comparable. They have superior performance than others but MPWR. It can be found that MPWR obviously outperforms the others involving both OA and κ. Moreover, when b is greater than 16, comparable or even higher accuracies than all bands are achieved by MPWR. This reflects that MPWR only eliminates the irrelevant, redundant, or noisy bands. Instead, the informative and distinctive bands are retained. To further give the performance evaluation, the classification accuracy curves versus b under LDA are shown in Fig. 8(b) and (e).

As shown in Fig. 8(b) and (e), VGBS exhibits apparent performance advantage over others for b greater than 11. The accuracies attained by VGBS are even higher than the ones with all the bands. However, when b is less than 11, VGBS is far inferior to WaLuDi, TRC-OC-FDPC, DSEBS, and MPWR. As demonstrated in Fig. 8(b) and (e), MPWR has comparable accuracies with WaLuDi and TRC-OC-FDPC. Furthermore, for b ranging from 1 to 2, the greatest accuracy improvement is still achieved by MPWR.

Concerning the EPF-based performance evaluation, Fig. 8(c) and (f) shows the accuracy curves against b of different methods. Analogous to Fig. 8(a) and (d), the superiority of MPWR is displayed intuitively. In particular,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

TABLE V

OA AND κ STATISTIC (IN PERCENTAGE) OBTAINED BY SVM, LDA, AND EPF FOR THE PAVIA UNIVERSITY IMAGE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Classifier	SVM								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	86.04	80.33	90.54	89.94	74.85	74.85	90.20	87.79	**93.03**
κ	77.02	69.92	84.67	84.20	59.78	59.78	84.58	80.47	**90.65**
Classifier	LDA								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	85.61	83.42	**90.35**	89.51	79.01	80.69	89.95	87.87	89.30
κ	80.44	79.57	**87.18**	85.43	73.64	75.97	86.12	83.69	85.61
Classifier	EPF								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	91.39	83.98	95.84	95.15	78.14	78.14	95.75	94.07	**99.11**
κ	81.72	73.37	90.15	89.60	62.79	62.79	90.59	86.73	**98.49**

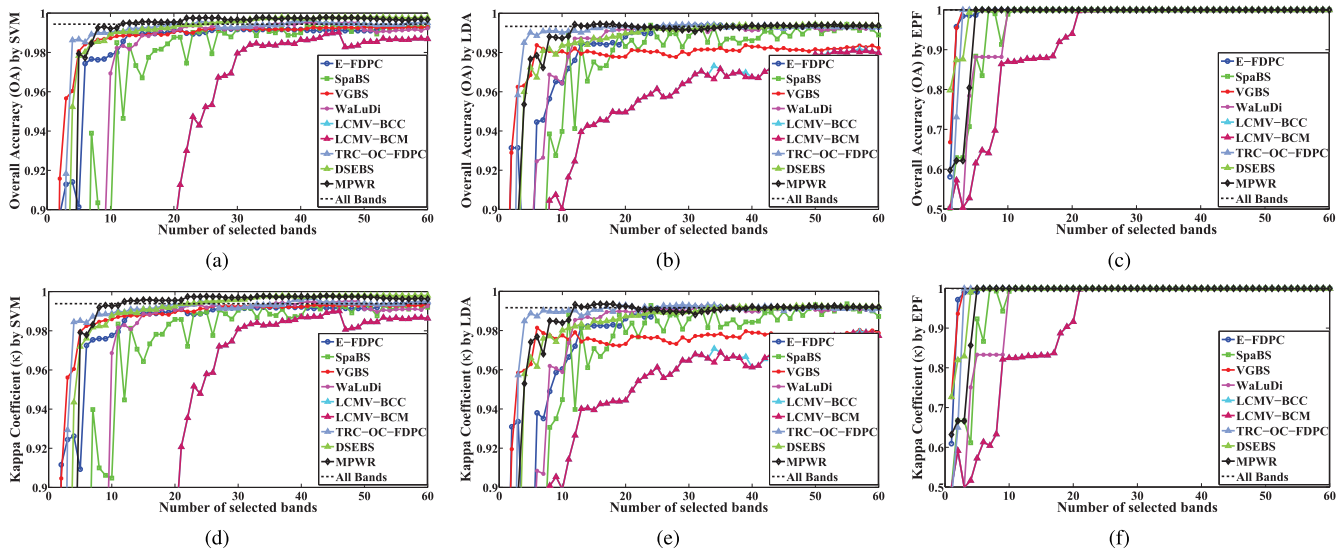


Fig. 9. Classification accuracy curves of different band selection methods on the Salinas-A image with different number of selected bands. OA obtained by (a) SVM, (b) LDA, and (c) EPF. κ obtained by (d) SVM, (e) LDA, and (f) EPF. The black dotted lines represent that all bands are used for classification without selection.

for b greater than 25, the accuracy curve of MPWR is prone to stabilize and gradually close to 1. For consistency, for b equal to 25, the specific accuracies generated by each method are also quantitatively presented in Table V.

G. Experimental Results on Salinas-A Image

This section provides the comparative experiments corresponding to different methods conducted on the Salinas-A image. In the experiments, around 10% of the Salinas-A images are randomly picked for training, and the remaining 90% are utilized for testing, as shown in Table III. Then, given the selected bands individually acquired by each band selection method, we can gain the classification accuracy through SVM, LDA, and EPF. For clarity and intuition, the SVM-based, LDA-based, and EPF-based OA values versus b are shown in Fig. 9(a)–(c), respectively. Accordingly, Fig. 9(d)–(f) demonstrates the SVM-based, LDA-based, and EPF-based κ curves against b.

Note that the Salinas-A image has six distinctive classes, which are easy to be distinguished. Therefore, as shown in Fig. 9, the accuracies on this image are higher than the ones on the Indian Pines and Pavia University images. It is evident that nine methods are competent to produce the OA values of over 0.9 in most cases. In this connection, to clearly show the comparison results, the y-axis varies from 0.9 to 1.

Despite the excellent classification results, there are still performance discrepancies for diverse band selection methods. As shown in Fig. 9(a) and (d), MPWR has obvious superiority over others for b greater than 8. For instance, for b equal to 8, compared with the lowest OA, MPWR actually improves the OA by over 0.08. Moreover, for b greater than 12, MPWR can always yield higher classification accuracies than all bands. This indicates that MPWR is capable of retaining the informative bands, whereas numerous noisy and redundant bands are discarded. For completeness, Table VI quantitatively lists the classwise classification accuracies for 12 selected bands as well, like Tables IV and V.

In terms of the LDA-based experimental results, MPWR still shows excellent classification performance. This further reveals that there exist numerous redundant or irrelevant spectral bands on this image, and 50 selected bands are qualified to attain a satisfying classification performance. The specific accuracy curves can be seen from Fig. 9(b) and (e).

In view of EPF, note that both the spectral and spatial structural information are introduced into classification. Therefore, EPF can produce better classification results as a whole. As depicted in Fig. 9(c) and (f), for b greater than 5, besides

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

OA AND κ STATISTIC (IN PERCENTAGE) OBTAINED BY SVM, LDA, AND EPF FOR THE SALINAS-A IMAGE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Classifier	SVM								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	98.56	94.64	98.85	98.33	79.85	79.85	99.06	99.04	**99.48**
κ	98.52	94.47	98.75	98.29	77.34	77.34	98.95	98.92	**99.50**
Classifier	LDA								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	97.60	94.12	98.21	98.71	92.46	92.46	99.10	98.48	**99.44**
κ	97.21	93.97	97.91	98.54	92.65	92.65	99.03	98.21	**99.33**
Classifier	EPF								
Band Selection Methods	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
OA	99.98	99.94	99.98	99.98	87.24	87.24	**100**	99.98	99.98
κ	99.95	99.90	99.95	99.95	82.67	82.67	**100**	99.95	99.95

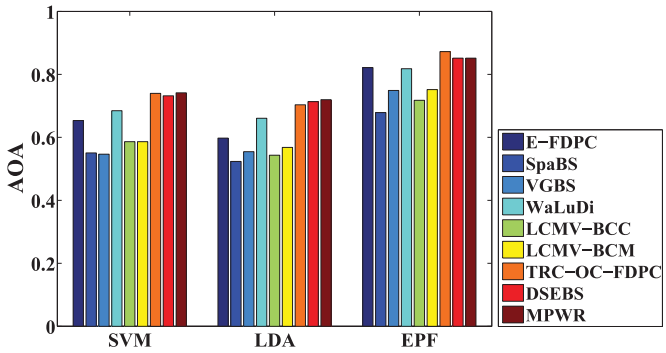


Fig. 10. Average of the OA values versus the first 60 selected bands on the Indian Pines image for different band selection methods.

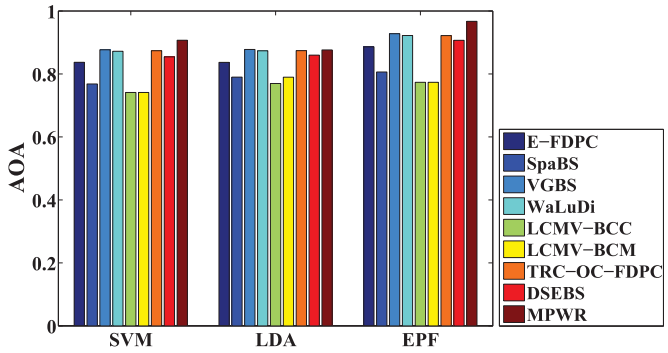


Fig. 11. Average of the OA values versus the first 50 selected bands on the Pavia University image for different band selection methods.

MPWR, TRC-OC-FDPF, DSEBS, and WaLuDi, VGBS also achieves the classification accuracies close to 1.

H. Average Overall Accuracy for Nine Methods

It is worth noting that with the variation of b, the performance ranking of nine methods is changing as well. For example, it can be seen from Fig. 7(a) that when b locates between 2 and 30, LCMV-BCM evidently achieve better performance than VGBS and SpaBS. However, for b greater than 40, SpaBS is superior than LCMV-BCM and VGBS. In view of this, the overall performance evaluation of each method on the three data sets should be investigated as well.

Therefore, for the Indian Pines image, Fig. 10 further presents the average performance of the first 60 bands with respect to each method. In this figure, the average OA values, denoted as AOA, against the first 60 selected bands for nine band selection methods are depicted. Each colored bar represents the AOA of the corresponding band selection method.

From the AOA depicted in Fig. 10, the merits and demerits of various methods are demonstrated intuitively. We can easily find that, in terms of both SVM-based AOA and LDA-based AOA, MPWR achieves the best performance. Additionally, concerning EPF-based AOA, MPWR is just inferior to TRC-OC-FDPC. Then, the excellence and robustness of MPWR is further reflected.

Analogously, on the Pavia University image, as b changes, an excellent method may exhibit performance degradation. Instead, a method with common performance could become excellent with the change of b. Take Fig. 6(b) as an example. It can be found that for b below 10, the performance of VGBS is not satisfying. Nonetheless, for b greater than 10, VGBS is outstanding and superior to others. Therefore, based on the local performance, it is difficult to determine which method is better. To this end, this section further gives the average performance of the first 50 bands attained by each method on the Pavia University image. For illustrative purpose, the specific values of AOA are shown in Fig. 11.

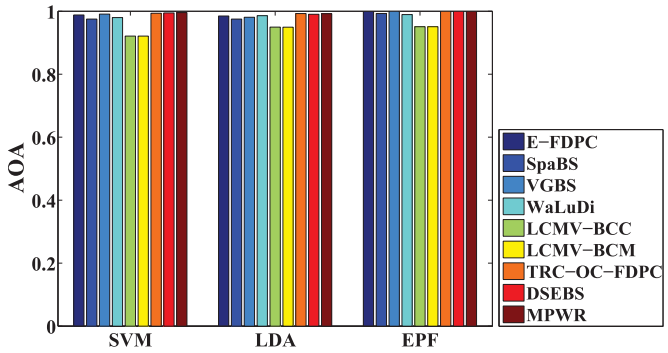


Fig. 12. Average of the OA values versus the first 60 selected bands on the Salinas-A scene for different band selection methods.

From Fig. 11, we can see that, considering the LDA-based AOA, VGBS outperforms MPWR. Regardless of SVM-based AOA or EPF-based AOA, the maximum AOA values are always yielded by MPWR. This clearly indicates the superiority and robustness of MPWR.

To evaluate the performance of each method as a whole, the average accuracies on Salinas-A image are also provided. Note that some results for b less than 5 are even about 0.3. This leads to great discrepancy with 0.9 in most cases. Therefore, the AOA shown in Fig. 12 corresponds to the b ranging from 5 to 60.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
RUNNING TIMES (S) OF EACH BAND SELECTION METHOD. THE LONGEST TIMES ARE HIGHLIGHTED IN BOLD TYPEFACE

Dataset	b	E-FDPC	SpaBS	VGBS	WaLuDi	LCMV-BCC	LCMV-BCM	TRC-OC-FDPC	DSEBS	MPWR
Indian Pines	12	0.0248	**3.8902**	0.1243	0.3230	0.0030	0.0024	0.6020	0.0259	0.2773
	20	0.0257	**5.6654**	0.1224	0.3264	0.0024	0.0032	0.7860	0.0279	0.2883
	30	0.0254	**8.8470**	0.1224	0.3248	0.0026	0.0034	0.9560	0.0265	0.2891
Pavia University	12	0.2657	**9.6593**	0.8012	1.1776	0.0036	0.0052	3.3510	0.0346	2.0095
	20	0.2630	**12.5026**	0.7915	1.1714	0.0034	0.0052	3.9084	0.0377	1.9893
	30	0.2774	**17.4074**	0.7994	1.1710	0.0035	0.0052	4.6270	0.0406	1.9911
Salinas-A	12	0.3498	**9.8597**	0.8170	1.1860	0.0082	0.0076	3.5146	0.0339	2.0470
	20	0.3369	**12.5321**	0.8150	1.1728	0.0098	0.0074	4.3270	0.0369	2.0690
	30	0.3478	**17.7276**	0.8233	1.1764	0.0081	0.0071	5.0140	0.0383	2.1060

Fig. 12 shows that on the Salinas-A image, most methods, e.g., E-FDPC, SpaBS, VGBS, and so on, achieve satisfying performance. In particular, the AOA values of them can reach over 0.9. While comparing these AOA values concretely, we can observe that MPWR gains the highest AOA. Then, the superiority and robustness of MPWR can be further demonstrated.

I. Computational Time and Space Complexity Analysis

As is known, computational time is an important aspect of embodying the merits and demerits of a method. In this connection, this section gives the run-time comparison of various band selection methods. Comparative experiments are conducted on a computer with Intel Core i7-6700k 4.00-GHz CPU and 32-GB RAM. In addition, all methods are implemented in MATLAB R2013b. Table VII lists the running times of each method for b equal to 12, 20, and 30, respectively. In this table, the execution times are measured in seconds. Moreover, considering the statistical significance of experimental results, all the running times are averaged over 20 independent runs.

According to Table VII, there exists no significant trend over b in view of the computational time of MPWR. Then, it can be inferred that the computational time of MPWR is not related to b. In contrast, TRC-OC-FDPC and SpaBS obviously take more time with the increase of b. From Table VII, we can find that LCMV-BCC and LCMV-BCM are the most efficient methods. Although MPWR is not as efficient as E-FDPC, WaLuDi, LCMV-BCM, LCMV-BCC, and DSEBS, it expends less execution time than TRC-OC-FDPC and SpaBS. Apparently, the computational time of MPWR is acceptable. Additionally, it is worth noting that MPWR has evident advantage over others regarding classification accuracies. Therefore, MPWR is capable of possessing both the acceptable efficiency and superior performance.

However, this section provides the contrast in the space complexity of diverse methods. Assume that the HSI contains n_r rows and n_c columns with l spectral bands. Then, the total number of pixels p is equal to $n_r n_c$. Table VIII depicts the space complexity comparison of each method.

As described in Table VIII, $O(lp + n_c p)$ space is needed for LCMV-BCC and LCMV-BCM. For SpaBS and VGBS, they tend to consume more space. Concretely, their space complexity is $O(l^2 + lp + n_c p)$. In comparison, E-FDPC, WaLuDi, TRC-OC-FDPC, DSEBS, and MPWR only require $O(l^2 + lp)$ space. As p is usually much greater than l, i.e., $l \ll p$, then $n_c p$ will exceed l^2 in most cases. In light of this, MPWR has the lowest spatial complexity as well.

TABLE VIII
SPACE REQUIREMENT OF EACH METHOD

Methods	Space complexity
E-FDPC	$O\left(l^2 + lp\right)$
SpaBS	$O\left(l^2 + lp + n_c p\right)$
VGBS	$O\left(l^2 + lp + n_c p\right)$
WaLuDi	$O\left(l^2 + lp\right)$
LCMV-BCC	$O\left(lp + n_c p\right)$
LCMV-BCM	$O\left(lp + n_c p\right)$
TRC-OC-FDPC	$O\left(l^2 + lp\right)$
DSEBS	$O\left(l^2 + lp\right)$
MPWR	$O\left(l^2 + lp\right)$

V. CONCLUSION

This article proposes an MPWR unsupervised band selection method. In this method, a constrained band-weight learning model is formulated. This ensures that both bandwise manifold-preserving capability and intraband correlation are simultaneously considered in the band selection process. In other words, the selected bands retain rich essential structural information, whereas they have low redundancy. Then, to solve the problem, an effective optimization algorithm on the basis of the ADMM framework is further developed. However, aimed at evaluating the bandwise manifold-preserving capability, a manifold-preserving band-importance metric $\boldsymbol{v}$ is devised. This metric enables us to identify the valuable bands involving abundant intrinsic geometric structural information. Moreover, for the sake of attaining $\boldsymbol{v}$, a corresponding attainment algorithm is presented. In specific, through embedding learning and linear regression, we obtain the coefficient matrix that characterizes the bandwise values to maintain the manifold structure. Then, the introduction of multi-normalization helps generate accurate probability-based $\boldsymbol{v}$. Concretely speaking, the columnwise maximum normalization is conductive to alleviate the magnitude discrepancy of the coefficient matrix and then contributes to acquiring $\boldsymbol{v}$ more accurately. In addition, through the employment of the min–max normalization, the probability-based $\boldsymbol{v}$, consistent with the intraband correlation in magnitude, can be yielded.

Comparative experiments, performed on the AVIRIS Indian Pines image, ROSIS Pavia University image, and AVIRIS Salinas-A image, demonstrate that though many methods are very effective, MPWR still exhibits superiority over the other six methods. Especially, the AOAs, shown in Figs. 10–12, indicate that MPWR is more robust and superior to others as a whole. Therefore, to explore effective and multi-scale features,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

in the future work, we would like to investigate the extraction of deep features instructed by MPWR.

ACKNOWLEDGMENT

The authors would like to thank Prof. D. Landgrebe from Purdue University and Prof. P. Gamba from Pavia University for providing the hyperspectral data set. They would also like to thank Z. Zhang for his work in the paper presentation.

REFERENCES

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[2] S. Zhang, J. Li, Z. Wu, and A. Plaza, "Spatial discontinuity-weighted sparse unmixing of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5767–5779, Oct. 2018.

[3] P. Shippert, "Why use hyperspectral imagery?" *Photogramm. Eng. Remote Sens.*, vol. 70, no. 4, pp. 377–396, Apr. 2004.

[4] J. Li, M. Khodadadzadeh, A. Plaza, X. Jia, and J. M. Bioucas-Dias, "A discontinuity preserving relaxation scheme for spectral-spatial hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 625–639, Feb. 2016.

[5] D. L. Donoho *et al.*, "High-dimensional data analysis: The curses and blessings of dimensionality," *AMS Math Challenges Lect.*, vol. 1, p. 32, Aug. 2000.

[6] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[7] Y.-J. Deng, H.-C. Li, K. Fu, Q. Du, and W. J. Emery, "Tensor low-rank discriminant embedding for hyperspectral image dimensionality reduction," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7183–7194, Dec. 2018.

[8] R. Hang and Q. Liu, "Dimensionality reduction of hyperspectral image using spatial regularized local graph discriminant embedding," *IEEE J. Sel. Topics Appl. Earth Observat. Remote Sens.*, vol. 11, no. 9, pp. 3262–3271, Sep. 2018.

[9] J. C. Harsanyi and C.-I. Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.

[10] A. Plaza, P. Martinez, J. Plaza, and R. Perez, "Dimensionality reduction and classification of hyperspectral image data using sequences of extended morphological transformations," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 466–479, Mar. 2005.

[11] M. A. Turk and A. P. Pentland, "Face recognition using eigenfaces," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 1991, pp. 586–591.

[12] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, 2000.

[13] W. Li, S. Prasad, J. E. Fowler, and L. M. Bruce, "Locality-preserving dimensionality reduction and classification for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1185–1198, Apr. 2012.

[14] D. Cai, X. He, J. Han, and H.-J. Zhang, "Orthogonal Laplacianfaces for face recognition," *IEEE Trans. Image Process.*, vol. 15, no. 11, pp. 3608–3614, Nov. 2006.

[15] M. Belkin and P. Niyogi, "Laplacian eigenmaps and spectral techniques for embedding and clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, 2001, vol. 14, no. 6, pp. 585–591.

[16] X. He, D. Cai, S. Yan, and H.-J. Zhang, "Neighborhood preserving embedding," in *Proc. 10th IEEE Int. Conf. Comput. Vis. (ICCV)*, vol. 2, Oct. 2005, pp. 1208–1213.

[17] P. N. Belhumeur, J. P. Hespanha, and D. Kriegman, "Eigenfaces vs. Fisherfaces: Recognition using class specific linear projection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 19, no. 7, pp. 711–720, Jul. 1997.

[18] K. Sun, X. Geng, and L. Ji, "A new sparsity-based band selection method for target detection of hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 2, pp. 329–333, Feb. 2015.

[19] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.

[20] W. Sun, L. Zhang, L. Zhang, and Y. M. Lai, "A dissimilarity-weighted sparse self-representation method for band selection in hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4374–4388, Sep. 2016.

[21] J.-M. Li and Y.-T. Qian, "Clustering-based hyperspectral band selection using sparse nonnegative matrix factorization," *J. Zhejiang Univ. Sci. C*, vol. 12, no. 7, pp. 542–549, 2011.

[22] H. Su, H. Yang, Q. Du, and Y. Sheng, "Semisupervised band clustering for dimensionality reduction of hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 6, pp. 1135–1139, Nov. 2011.

[23] J. Bai, S. Xiang, and C. Pan, "Classification oriented semi-supervised band selection for hyperspectral images," in *Proc. IEEE Int. Conf. Pattern Recognit. (ICPR)*, Nov. 2012, pp. 1888–1891.

[24] X. Bai, Z. Guo, Y. Wang, Z. Zhang, and J. Zhou, "Semisupervised hyperspectral band selection via spectral–spatial hypergraph model," *IEEE J. Sel. Topics Appl. Earth Observat. Remote Sens.*, vol. 8, no. 6, pp. 2774–2783, Jun. 2015.

[25] S. Li and H. Qi, "Sparse representation based band selection for hyperspectral images," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2011, pp. 2693–2696.

[26] C. Sui, Y. Tian, Y. Xu, and Y. Xie, "Unsupervised band selection by integrating the overall accuracy and redundancy," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 185–189, Jan. 2015.

[27] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[28] Z. Guo, X. Bai, Z. Zhang, and J. Zhou, "A hypergraph based semi-supervised band selection method for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2013, pp. 3137–3141.

[29] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

[30] Y. Yuan, X. Zheng, and X. Lu, "Discovering diverse subset for unsupervised hyperspectral band selection," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 51–64, Jan. 2017.

[31] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[32] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016.

[33] A. Martínez-Usó, F. Pla, J. M. Sotoca, and P. García-Sevilla, "Clustering-based hyperspectral band selection using information measures," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4158–4171, Dec. 2007.

[34] C.-I. Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.

[35] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[36] S. L. Moan, A. Mansouri, Y. Voisin, and J. Y. Hardeberg, "A constrained band selection method based on information measures for spectral image color visualization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 5104–5115, Dec. 2011.

[37] Z. Xue, P. Du, J. Li, and H. Su, "Simultaneous sparse graph embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6114–6133, Nov. 2015.

[38] L. Ma, M. M. Crawford, X. Yang, and Y. Guo, "Local-manifold-learning-based graph construction for semisupervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2832–2844, May 2015.

[39] Z. Fei, A. Halimi, P. Honeine, B. Chen, and N. Zheng, "Correntropy maximization via ADMM: Application to robust hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 4944–4955, Sep. 2017.

[40] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2011.

[41] G. Zhu, Y. Huang, J. Lei, Z. Bi, and F. Xu, "Unsupervised hyperspectral band selection by dominant set extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 227–239, Jan. 2016.

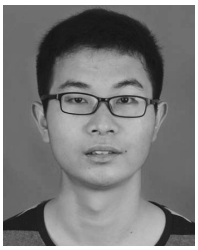
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.

[42] X. Geng, K. Sun, L. Ji, and Y. Zhao, "A fast volume-gradient-based band selection method for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7111–7119, Nov. 2014.
[43] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.
[44] M. Belkin, P. Niyogi, V. Sindhwani, and P. Bartlett, "Manifold regularization: A geometric framework for learning from labeled and unlabeled examples," *J. Mach. Learn. Res.*, vol. 7, no. 1, pp. 2399–2434, 2006.
[45] F. Nie, X. Wang, M. I. Jordan, and H. Huang, "The constrained Laplacian rank algorithm for graph-based clustering," in *Proc. 30th AAAI Conf. Artif. Intell.*, 2016, pp. 1–8.
[46] M. Wu and B. Scholkopf, "Transductive classification via local learning regularization," in *Proc. IEEE 11th Int. Conf. Artif. Intell. Statist.*, Mar. 2007, pp. 628–635.
[47] B. Cheng, J. Yang, S. Yan, Y. Fu, and T. S. Huang, "Learning with l^1-graph for image analysis," *IEEE Trans. Image Process.*, vol. 19, no. 4, pp. 858–866, Apr. 2010.
[48] X. He, S. Yan, Y. Hu, P. Niyogi, and H.-J. Zhang, "Face recognition using Laplacianfaces," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 3, pp. 328–340, Mar. 2005.
[49] P. Ahlgren, B. Jarneving, and R. Rousseau, "Requirements for a cocitation similarity measure, with special reference to Pearson's correlation coefficient," *J. Amer. Soc. Inf. Sci. Technol.*, vol. 54, no. 6, pp. 550–560, 2003.
[50] J. Benesty, J. Chen, and Y. Huang, "On the importance of the pearson correlation coefficient in noise reduction," *IEEE Trans. Audio, Speech, Language Process.*, vol. 16, no. 4, pp. 757–765, May 2008.
[51] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J. Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.
[52] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.
[53] H. Su, B. Zhao, Q. Du, and P. Du, "Kernel collaborative representation with local correlation features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1230–1241, Feb. 2018.
[54] Y. Gu, J. Chanussot, X. Jia, and J. A. Benediktsson, "Multiple Kernel learning for hyperspectral image classification: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6547–6565, Nov. 2017.
[55] Y. Bazi and F. Melgani, "Toward an optimal SVM classification system for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3374–3385, Nov. 2006.
[56] W. Sun, C. Liu, Y. Xu, L. Tian, and W. Li, "A band-weighted support vector machine method for hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1710–1714, Oct. 2017.
[57] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.
[58] X. Kang, S. Li, and J. A. Benediktsson, "Spectral-spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.



Chenhong Sui (M'19) received the Ph.D. degree in spatial information science and technology from the Huazhong University of Science and Technology, Wuhan, China, in 2015.

She is currently an Associate Professor with the School of Opto-Electronic Information Science and Technology, Yantai University, Yantai, China. Her research interests include remote sensing image processing, pattern recognition, and image quality assessment.



Chang Li received the B.S. degree in information and computing science from the Wuhan Institute of Technology, Wuhan, China, in 2012, and the Ph.D. degree in circuits and systems from the School of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan, in 2018.

He is currently a Lecturer with the Department of Biomedical Engineering, Hefei University of Technology, Hefei, China. His research interests include biomedical signal processing, hyperspectral image analysis, computer vision, pattern recognition, and machine learning.



Jie Feng (M'15) received the B.S. degree in electronic and information engineering from Chang'an University, Xi'an, China, in 2008, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2014.

She is currently an Associate Professor with the Laboratory of Intelligent Perception and Image Understanding, Xidian University. Her research interests include remote sensing image processing, deep learning, and machine learning.



Xiaoguang Mei received the B.S. degree in communication engineering from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 2007, the M.S. degree in communications and information systems from Huazhong Normal University, Wuhan, in 2011, and the Ph.D. degree in circuits and systems from HUST in 2016.

From 2010 to 2012, he was a Software Engineer with the 722 Research Institute, China Shipbuilding Industry Corporation, Wuhan. He is currently an Assistant Professor with the Electronic Information School, Wuhan University, Wuhan. His research interests include hyperspectral imagery, machine learning, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:26:10 UTC from IEEE Xplore. Restrictions apply.